SEEDLINE · FEASIBILITY ASSESSMENT

SeedLine feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 pipeline, kit 1.9.1 upgrade), for the current design (option A), closed at stage P3

SeedLine feasibility assessment

SeedLine is feasible to build and to work on paper: every part can be made with a saw, drill, taps, spanners and a desktop 3D printer from stock sold in regional towns, and the drive, metering, structure and work rate check out at first order (SDL-STM-001, SDL-CAL-001). The open feasibility questions are performance ones that only a bench or field test can settle at TRL 4: whether the printed cells fill reliably at the walking-speed rule, the miss and double rates behind the spacing uniformity, and the push force on real soils. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design (working principle)	A ground wheel driving a cell plate through a chain is the established principle of plate seeders; spacing follows travel with 1.2 to 1.7 % skid; 22 to 589 mm spacing range with the ratio kit	High	Cell fill at 0.299 m/s against an assumed 0.30 m/s limit (STM S6); spacing uniformity set by assumed miss and double rates (S7)	SDL-STM-001 S1 to S7; SDL-CAL-001 sections 2, 3, 6, 10
Design (operator)	Push 132 N horizontal in the design case, 14.8 kg with the marker kit, grip 850 to 1,050 mm	Medium	Push on heavy, loose seedbeds about 179 N; sustained pushing over a planting day not yet assessed (S9, SDL-REG-001 G5); mass margin 0.18 kg	SDL-CAL-001 sections 5, 6; SDL-STM-001 S8 to S10
Materials	Painted mild steel tube, plate and flat bar; PETG prints (ASA in sun); bought wheels, bearings, #35 chain and sprockets; aluminium tube for the optional marker arm	High	PETG creeps and ages in sun (S20); plate wear life unknown (PR-015); treated seed contact with plastics not assessed	SDL-STM-001 S19, S20; `bom/bom.csv`
Manufacturing and assembly	Cut, drill, tap, bend and bolt; no welding; housing printed whole with supports; 24 making sketches and a 17-step build plan; every part checked for fit in the kit 1.9 model (98 constructability checks and the CAD fidelity check)	High	Four fit details found by the detailed model and fixed (SDL-DEC-001 F1 to F4); tapped M5 and M6 holes in 3 mm plate are shallow (use inserts or nuts if they strip)	SDL-BLD-001; `cad/fidelity.json` ; `python cad/src/model.py --check`
Power and controls	Human-powered; no electronics; spacing set by plate and sprocket choice	High	None at TRL 3	SDL-PRC-001
Supply of parts	Steel sections, bearings, trolley wheels and roller chain are common in regional towns; desktop printers are in towns, universities and maker spaces, rare in villages	Medium	Hubbed #35 sprockets and a go-kart chain tensioner may need ordering; who prints plates is open (SDL-PRB-001 open questions)	`bom/bom.csv` ; SDL-MKT-001
Cost	USD 224.10 base, USD 266.10 with both kits, against value-engineering targets of about USD 200 and USD 400	Medium	Base seeder USD 24.10 over its target; savings listed in SDL-DEC-001	SDL-CAL-001 section 11

Ratings: High, Medium, Low.

Feasibility is closed at P3 for paper work: no aspect is rated Low. The Medium ratings carry into the risk register (SDL-RSK-001) and the TRL 4 plan on hold (SDL-PLN-001).